postlab report 1 - DSP

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April 2026

Lab 1: Adaptive Filtering

Part 1 : System analysis

a) parameter calculation

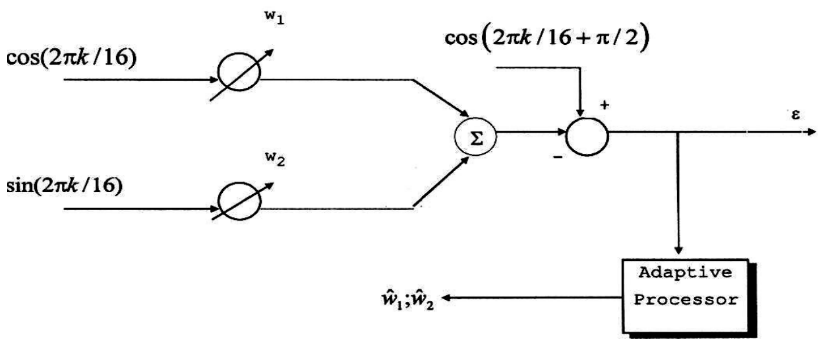

Figure 1: given system figure 2.1

- The output of the given system is derived to be

$$Y_k = w_1 \cdot cos(2\pi k/16) + w_2 \cdot sin(2\pi k/16)$$

- The error of the system is derived to be

$$\varepsilon_k = D_k - Y_k = cos(2\pi k/16 + \pi/2) - [w_1 \cdot cos(2\pi k/16) + w_2 \cdot sin(2\pi k/16)]$$

$$\{cos(\alpha + \pi/2) = -sin(\alpha)\} \Longrightarrow \varepsilon_k = -[w_1 \cdot cos(2\pi k/16) + (1 + w_2) \cdot sin(2\pi k/16)]$$

- It can be easily seen that for the choice of $w_1 = 0 \quad w_2 = -1$ the error is exactly 0 meaning full accuracy of the system.
- Now we will calculate the weights theoretically using the performance function derived in the prelab report

$$\zeta_k = MSE = \mathbb{E}[d_k^2] + W^T R W - 2W^T P$$

- We can easily substitute the matrices P and R based of the given system as follows

$$\zeta_k = \frac{1}{2} + W^T I_2 W - 2W^T \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\zeta_k = \frac{1}{2}[w_1^2 + w_2^2 + 2w_2 + 1]$$

$$\nabla_W \zeta_k = \frac{1}{2} \begin{bmatrix} 2w_1 \\ 2w_2 + 2 \end{bmatrix} = 0 \Longrightarrow W^* = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

- As expected we got the optimal solution calculated earlier. The fact that the signals are deterministic did not affect the correctness of the calculation process according to theory.

- the condition for the convergence of the algorithm, given that $R = \frac{1}{2}I_2 \Longleftrightarrow \lambda_{max} = \frac{1}{2}$ is

$$0 < \mu < \frac{1}{\lambda_{max}} \Longrightarrow 0 < \mu < 1$$

b) numerical analysis of system parameters

- first lets calculate the expected theoretical results of the algorithm in runtime. since the signals are deterministic, we expect that they algorithm will converge to the reference signal at full.

- for the analysis of the step size μ we want to show that :
  1) the MSE converges for step sizes satisfying the convergence condition
  2) the weights converge to the theoretical value calculated

- To show the analysis of the algorithm with respect to the step sizes we will run it with different sizes and analyze the results.

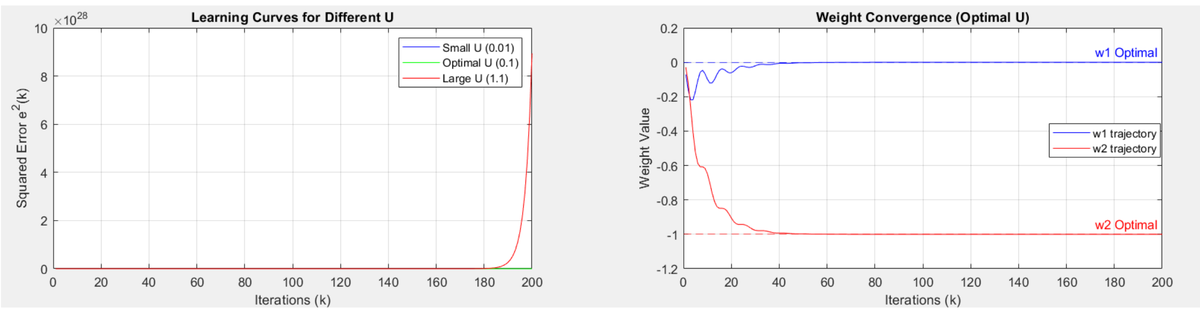

Figure 2: Running SD_LMS for a given W for different step sizes

- If we look closely at the left plot inside figure 2 we can see that the algorithm converges for all μ values that satisfy the condition $0 < \mu < \frac{1}{\lambda_{max}} = 1$ and diverge otherwise, as expected.

- And if we look at the right plot we can see that indeed that algorithm converges to the correct values of W^* which is reassuring.

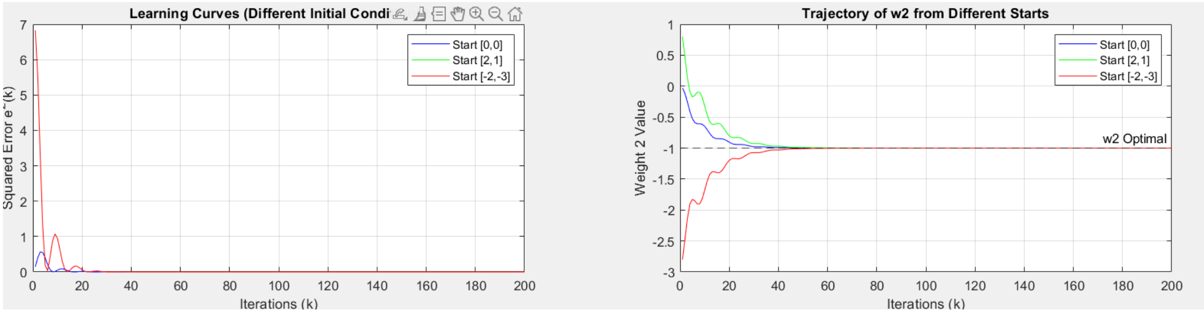

Figure 3: Running SD_LMS for a given μ for different initial conditions W_0

- If we look closely at the left plot inside figure 2 we can see that the algorithm converges for all W_0, as expected since the convergence condition does not depend on the initial conditions.
- And if we look at the right plot we can see that indeed that algorithm converges to the correct values of W^* for all initial conditions.

c) Comparing between numerical and theoretical analysis

```
Excess_MSE_exp = mean(Er_o(end-50:end).^2);
```

```
>> Excess_MSE_exp

Excess_MSE_exp =

   1.6365e-16

fx >>
```

Figure 4: numerical calculation of excess MSE based on runs performed in 1.b) taken from the last 50 samples after the weights are converged

1. **Excess-MSE :** After calculating the excess MSE straight from results we can see that it is practically 0. That is expected since the signals are deterministic and there should not be any error after convergence of the weights.
2. **Misadjustment :** When trying to calculate the Misadjustment analytically we need to be careful not to simply calculate $M = \frac{EMSE}{\zeta_{min}} = \frac{0}{0}?$. It is an incorrect approach... what we will do instead is look at the limit as $n \longrightarrow \infty$ of the two quantities. According to derivation in the book, for deterministic signals, that is derived to be

$$M \approx \mu \cdot tr(R) = \mu \sum_{i=1}^{N} \lambda_i$$

which for our case is

$$R = \frac{1}{2} I_2 \Longrightarrow M \approx \mu \cdot tr(\frac{1}{2} I_2) = \mu$$

*But in practice the numerical precision of matlab will not show that so it is basically N/A.

3. τ_{MSE}, T_{MSE} :

The theoretical value of the time constants are given by

```
--- Time Constants (Numerical vs Analytical for U=0.1) ---
Experimental tau_MSE: 5 iterations
Analytical tau_MSE:   5 iterations
Experimental T_MSE:   19 iterations
Analytical T_MSE:     20 iterations
```

Figure 5: numerical calculation of time constants

$$\tau_{MSE} = \frac{1}{4\mu\lambda} = \{\lambda_1 = \lambda_2 = 0.5\} = \frac{1}{2\mu}$$
$$T_{MSE} = 4\tau_{MSE} = \frac{1}{\mu\lambda} = \frac{2}{\mu}$$

meaning that for a value of $\mu = 0.1$, the MSE should drop to about 37% of its distance from the minimal MSE after about 5 iterations of the algorithm and to about 2% after 20 iterations.
That can be seen in the learning curve in figure 1 and 2 for the leanring curves of $\mu = 0.1$ and in figure 5.

4. ΔW **- distance from optimal solution :**

```
>> Weight_Error_Sq(end)

ans =

   4.3251e-20
```

Figure 6: last value of distance from optimal solution

Since the signals are deterministic, the weights are to converge to the optimal solution, meaning distance of 0. Figure 6 shows the distance between the theory and practice in the last iteration of the algorithm which is practically 0, as expected.

Part 2 : Active Noise Cancellation (ANC)

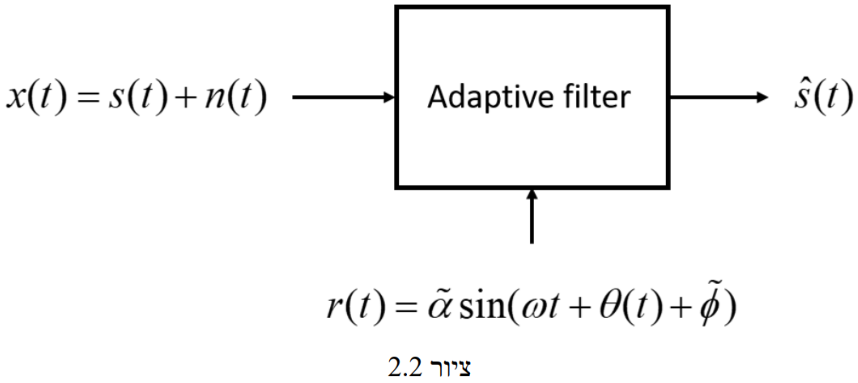

Figure 7: Given system diagram

a) + b) comparison between noisy and denoised sound track

after following the assignment instructions and filtering out the noise, we can hear the speech in the soundtrack

```
% 6. Playing noisy audio
sound(x, Fs);
```

```
% Playing denoised audio
sound(s_hat, Fs);
% The sound track says : "dont ask me to carry an oiley rag like that"
```

Figure 8: code running both soundtrack and discovering what it says

c) Visual analysis of PSD's

In this section we are asked to display the reconstruction of the original signal. If we look closely at the bottom and top plots in figure 9, we can see the very noticeable similarity between the two. Beside the sound we heard previously, this further proves that the reconstruction is successful. the periodic spikes we see in the middle and bottom plots are the residues of the predefined noise we added which is a triangular wave. Such a wave will have harmonies in same distance intervals.

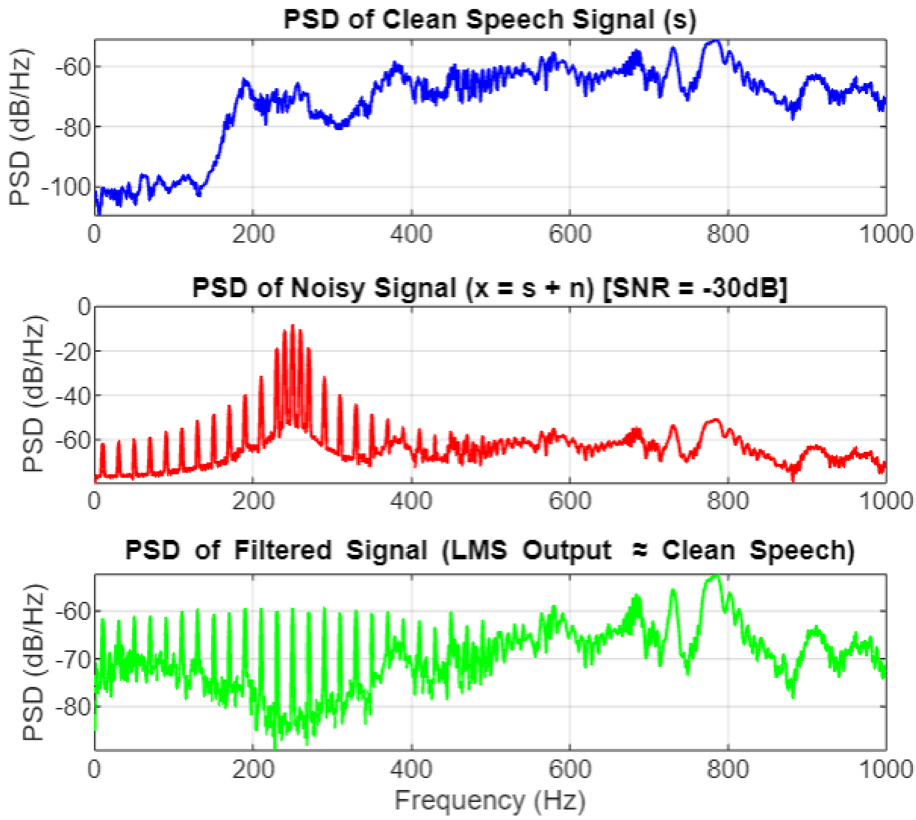

Figure 9: Reconstruction of original signal with using SD-LMS

d) Numerical analysis of parameters : μ, L

In this section we tested different values of μ, L the step size and the number of tap inputs accordingly in attempt to see their influence on the learning curve.

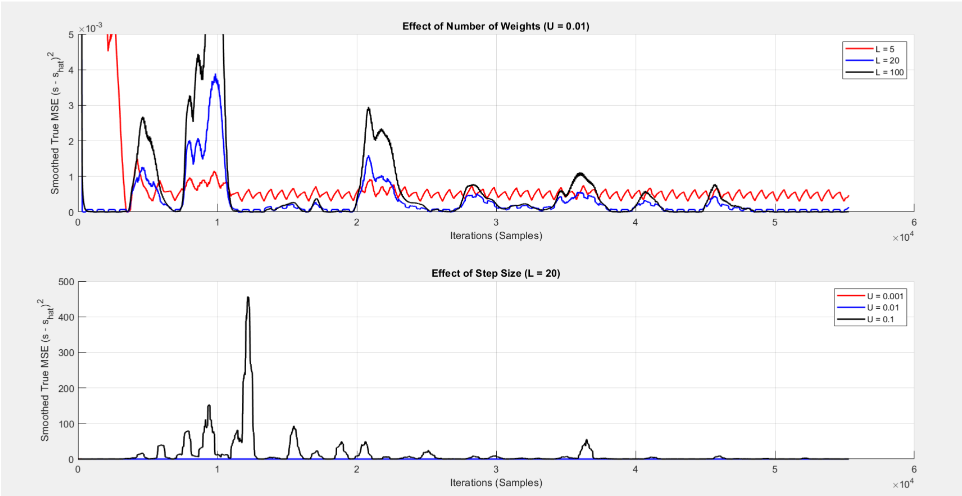

Figure 10: Reconstruction of original signal with using SD-LMS

- In the bottom plot of figure 10, we compare between different values of μ. The tradeoff between convergence rate and accuracy is visualized by this plot.

  - Too high and the MSE bounces up even after a lot of iterations.
  - Too low and the MSE is so high the plot is not even visable in the current scale .
  - The value in between is closer to the "sweet spot".

- In the top plot of figure 10, we compare between different values of L. The tradeoff between convergence rate and accuracy is ,yet again, visualized by the plot.

  - Too high and the MSE bounces up even after a lot of iterations.
  - Too low and the MSE reaches a minimum higher than potentially obtainable .
  - The value in between is closer to the "sweet spot".

e) Contrast with regular filters

To cancel the main 250 Hz spike and all of its 10 Hz harmonics with a regular filter, you would have to meticulously design a highly complex "comb" filter with hundreds of perfectly placed notches. If the noise properties (like the 10 Hz modulation rate) drift even slightly in the real world, your precisely designed fixed filter instantly becomes useless.

Part 3 : FIR system recognition

a) Synthesize system output

Here are the results simulated in matlab displayed in figure 11

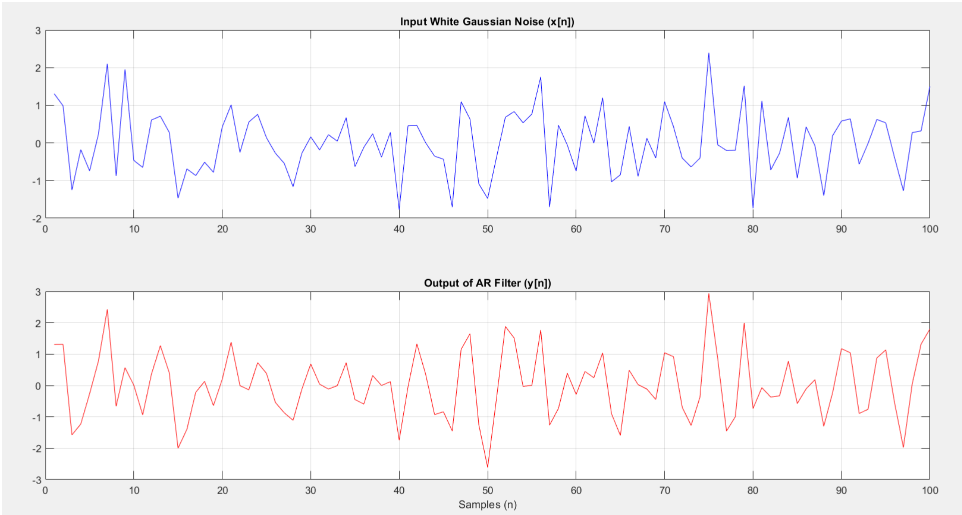

Figure 11: Input noise and synthesized output signal, first 100 samples

b) Numerical analysis of parameters : L, N, μ

1. **L - number of weights:** We trying to model an Infinite Impulse Response (IIR) system using a Finite Impulse Response (FIR) filter. An IIR filter's true impulse response is technically infinite in length. Therefore, the implication is that we need a large number of FIR weights to adequately capture the long, decaying tail of the IIR system's impulse response. If we use too few weights, we truncate the response, leading to poor modeling and high steady-state error.

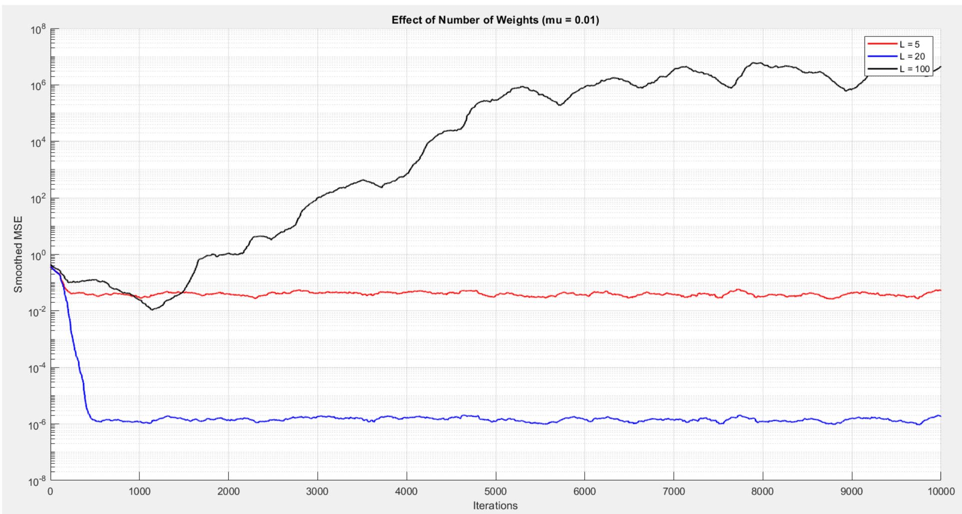

Figure 12: Influence of number of weights

2. **N - number of iterations :** As usual compared to previous exercises, the number of iterations further trains the weights and lowers the MSE until reached steady-state where the MSE is minimal for that specific learning process.

3. **μ - step size :** In figure 13 we can see the same tradeoff discussed in previous parts of the report. we can point out that for $\mu = 0.1$ the step size doesn't satisfy the convergence condition and very visibly diverges.

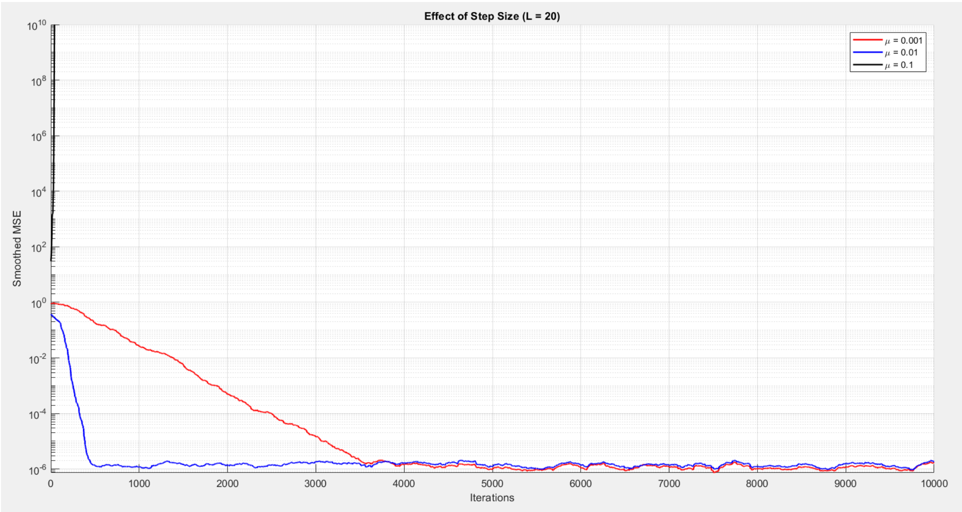

Figure 13: Influence of step size

initial conditions : here we can see the final weights after starting from 0 and after starting from some random value. Both land on the same value, therefore, the training results do not depend on initial conditions.

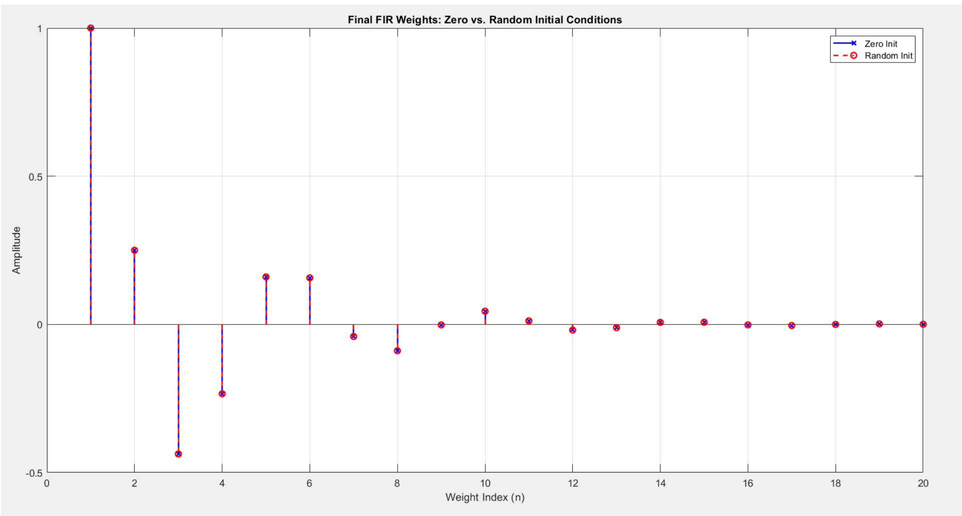

Figure 14: blue weights initialized at 0, red weights initialized at random values

convergence to true value In figure 14 we can see the final weights compared with the true value given in the beginning of the exercise.

```
--- Original System Coefficients ---
a1 = -0.25, a2 = 0.5
--- Estimated Coefficients (Zero Init) ---
a1_est = -0.2501
a2_est = 0.4999
```

Figure 15: Final values of weights

c) original filter vs estimated FIR

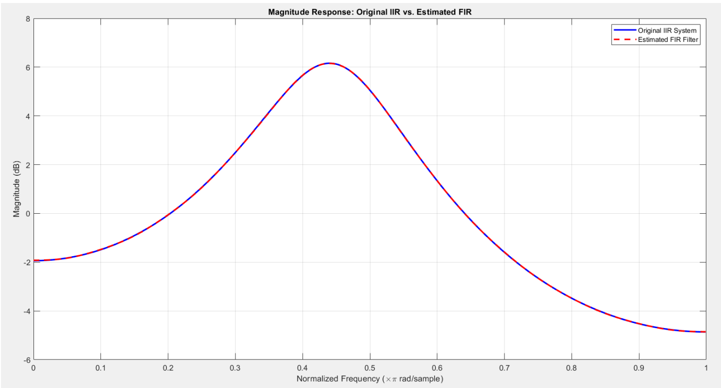

Figure 16: The filters are aligned perfect to the naked eye showing a successful reconstruction of the transfer function